

The Link Function Problem in Psychological Interaction Testing

Margherita Calderan¹, Filippo Gambarota², Laura Sità¹, Tommaso Feraco¹, and Enrico Toffalini¹

¹Department of General Psychology, University of Padova

²Department of Developmental Psychology and Socialization, University of Padova

Author Note

Margherita Calderan  <https://orcid.org/0009-0005-5668-5162>

Filippo Gambarota  <https://orcid.org/0000-0002-6666-1747>

Laura Sità  <https://orcid.org/0009-0000-6017-9979>

Tommaso Feraco  <https://orcid.org/0000-0002-8920-5330>

Enrico Toffalini  <https://orcid.org/0000-0002-1404-5133>

Correspondence concerning this article should be addressed to Enrico Toffalini,
Department of General Psychology, University of Padova, Via Venezia 8, Padova 35131, Italy,
Email: enrico.toffalini@unipd.it

Abstract

Equal differences on one scale become different differences on another scale. This is crucial for interaction testing because an interaction is a difference between differences, and in psychological research the observed response and the construct of interest are often expressed on different scales. Therefore, the same data can show an interaction on one scale and not on another.

Generalized Linear Models make this problem explicit, as the link function defines the scale on which the model without an interaction term is additive. Choosing a link thus defines what ‘no interaction’ means, while the family describes the nature of the response variable. The paper has two connected parts. First, a preregistered review of recent articles from five leading psychology journals shows that interaction testing is routinely performed in research, that the link function is most often unspecified, and that constrained outcomes are routinely analyzed with identity-link models without justifying the assumption of equal effects on the observed scale. Second, worked examples and simulations develop three common response formats: forced-choice accuracy with a chance floor, within-family link choices (logit vs probit), and sum scores as a measurement case, and quantify how a plausible-looking link can turn an additive structure on one scale into a statistically significant interaction on another. A final diagnostic simulation shows that residual checks, a link test, and AIC comparisons give uneven protection and do not by themselves decide which scale the scientific claim concerns. Interaction claims remain underspecified until researchers state, justify, and probe the scale on which no interaction is assumed.

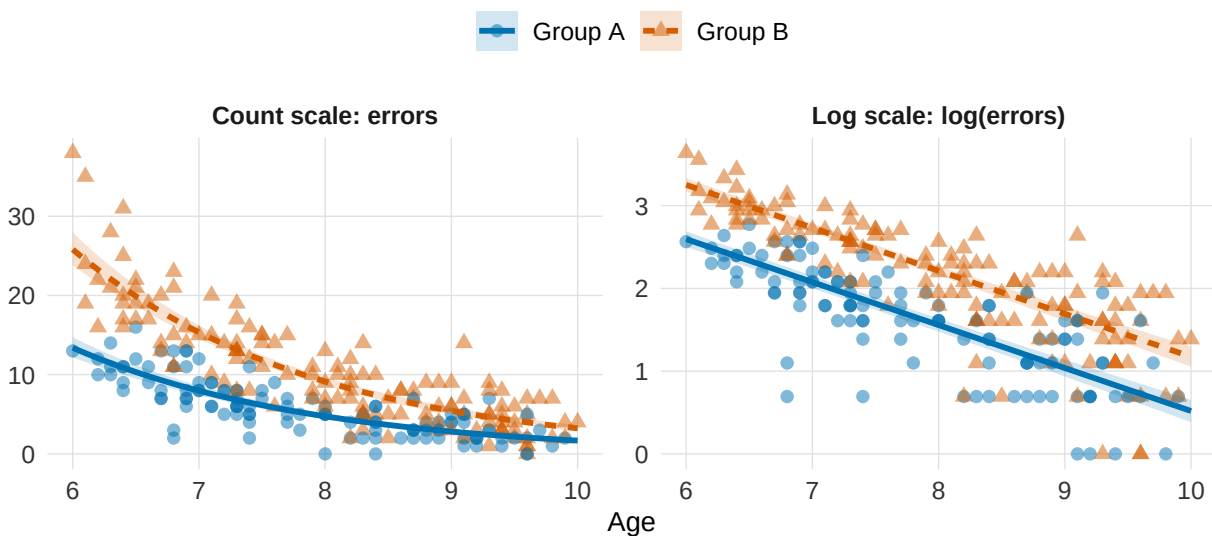
Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

We start with a very simplified example. Two populations of children (e.g., clinical vs neurotypical) differ in their overall error rates on a task, and errors decrease with age in both. Figure 1 samples simulated data from such a scenario. In the left panel (observed response), errors decline with age in both groups, and the difference between groups is much larger at age six than at age ten. Is there an age-by-group interaction here?

Figure 1

Simulated error counts for two groups of children aged 6 to 10, shown on the observed count scale (left) and on the log scale (right). The two groups have exactly the same age effect on the log of the expected error count, so on the log scale the two fitted lines are parallel, with a constant group gap at every age. Mapped back onto the count scale, the same model produces converging curves, so the group difference shrinks with age. Whether this pattern is an interaction depends on the scale on which effects are assumed to add: on the count scale the group difference changes with age; on the log scale it does not.



Because an interaction is a difference between differences, the question is whether the group difference itself changes with age. On the observed count scale, the group effect does appear to shrink as age increases, so it looks like an interaction. But looking at the same data on

the log-scale (Figure 1, right panel), both groups have the same age slope, meaning their fitted lines are parallel. A constant difference on the log scale corresponds to a constant ratio on the count scale, so the absolute gap between groups is proportional to the baseline and shrinks as errors decline with age. Whether we call this an interaction is not a fact about the data, it depends on which scale we assume effects add up linearly on. Additive on the log scale means multiplicative (and non-parallel) on the count scale, and vice versa.

This ambiguity extends far beyond this single illustration, as interaction hypotheses are widespread in psychological research and often represent the central target of inquiry (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021). Whenever researchers ask whether an effect is moderated, or whether it holds only under certain boundary conditions, they are asserting that the impact of one variable depends on another. This count example therefore demonstrates that the reality of such a claim can shift entirely based on the scale on which those differences are computed, as equal differences on one scale are different differences on another scale.

In psychology, this scale dependence has a characteristic source in measurement. Researchers routinely conceptualize constructs such as cognitive ability or symptom severity as continuous latent dimensions (Embretson & Reise, 2000; Wagenmakers et al., 2012). Those dimensions are rarely observed directly; what reaches the data are bounded, discrete quantities, such as counts, ratings, and accuracy scores that carry floors, ceilings, and chance levels. Each measurement scale imposes its own mapping from the latent dimension to the observed response. Where the mapping compresses, a constant effect on the latent dimension appears as a shrinking effect on the observed response: a group or condition closer to a bound has less room to move, and can look less affected by a manipulation, than a group or condition farther away from bounds, even if the underlying data-generating process remains the same at all levels of the scale.

Generalized linear models (GLMs) make this mapping explicit and give it a standard vocabulary (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972), separating two choices that are easy to conflate: the outcome family, which describes the response, and the link function, which sets the scale on which effects combine (Box 1).

Box 1: Generalized linear models: outcome family and link function

The *outcome family* is the distribution assumed for the response. It defines which values the outcome can take and how observations vary around their expectation. The *link function*, $g(\cdot)$, maps the expected outcome, $\mu_i = E(Y_i)$, onto the linear predictor, η_i , the scale on which the model is additive. With two predictors and their interaction term,

$$g(\mu_i) = \eta_i, \quad \eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The inverse link, $g^{-1}(\cdot)$, maps this additive structure back onto the expected response. Its familiar role is to keep expected values within the admissible range of a constrained outcome: $\exp(\eta)$ keeps expected counts positive, and the inverse logit keeps probabilities between 0 and 1. Less obviously, this also fixes what the absence of an interaction, $\beta_3 = 0$, means in the fitted model. The product-term coefficient β_3 is therefore an interaction on the link scale. A response-scale interaction is a different quantity: a difference between expected differences on the observed outcome, for example, whether the group difference in expected errors changes with age in Figure 1. Unless the link is the identity, the two quantities can disagree (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Each family has a canonical link, but other links are legitimate. A Gaussian linear model is the special case that combines a Gaussian family with an identity link, so the additive scale and the response scale coincide and the scale question is invisible. The model behind Figure 1 is a Poisson family with a log link. The family respects that errors are non-negative counts, and the link places additivity on the log scale.

Often the outcome family is chosen with care while the link function is left at its software default, fixing the scale of additivity without deliberation. Two models can share a family and still place additivity on different scales. For example, a Gamma family suits reaction times data; within that family, a log link makes effects additive in log milliseconds, a proportional change, and the canonical inverse link makes them additive in reciprocal means, a change in speed. A

binomial family fits correct-versus-incorrect responses and leaves the link open among logit, probit, chance-corrected, and other forms.

The scale dependence of interactions has long been recognized. Loftus showed that interactions involving response probabilities are meaningful only relative to a mapping between observed and theoretical scales (Loftus, 1978). Cox formalized interaction as a model-relative concept tied to transformation and interpretation (Cox, 1984). Wagenmakers and colleagues distinguished removable from nonremovable interactions, showing that some depend entirely on the measurement scale (Wagenmakers et al., 2012). Busemeyer showed that conclusions about combination rules depend on the measurement model (Busemeyer & Jones, 1983), and Lubinski and Humphreys illustrated how model specification choices can mimic moderator effects (Lubinski & Humphreys, 1990). More recently, Rohrer and Arslan argued that interaction tests often provide precise answers to vague questions when the estimand and scale are left unspecified (Rohrer & Arslan, 2021), and Rimpler and colleagues showed that product terms in linear models can mismatch the functional form implied by a substantive conditional-effect claim (Rimpler et al., 2026). In GLMs specifically, product-term coefficients are not generally equal to response-scale interactions, so interaction effects should be evaluated as marginal effects or response-scale contrasts (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Domingue and colleagues demonstrated that outcome-model misspecification can produce false interaction findings for binary, count, and censored outcomes (Domingue et al., 2024), and Liddell and Kruschke showed that treating ordinal responses as metric can inflate error rates for interactions (Liddell & Kruschke, 2018). Czado and Santner, and later Yu and Huang, studied the inferential consequences of link misspecification in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019). Interaction estimates are also fragile for ordinary reasons, including small effects, low reliability, and limited power (Baranger et al., 2023; Castillo et al., 2026).

These contributions identify adjacent parts of a shared problem. We build on them and take up the case in which the outcome family is already broadly plausible and the link alone sets a disputable no-interaction baseline, which is the common situation for bounded and chance-limited

psychological outcomes. An interaction on a scale other than the generating one is not a false positive in a strict statistical sense; it is a genuine mathematical property of that scale (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022). We define a pseudo-interaction as an interaction contrast that is zero on the known generating scale but nonzero on another fitted scale because the two scales encode different additivity baselines. This label is available in simulations because the generating process is known. In empirical data, where the generating scale is unknown, we instead describe interaction conclusions as link-sensitive, conditional on the fitted scale, or unstable across plausible links. Some interactions survive any monotonic rescaling, crossover interactions being the classical example (Wagenmakers et al., 2012). Many others do not, and the present paper focuses on these: the incremental contribution is to separate outcome family, link function, and target metric in a single framework for interaction testing, combine this separation with a preregistered review documenting the reporting gap, and quantify the inferential risk through simulations of common psychological response formats.

The present article addresses these issues in two parts. The first is a preregistered descriptive review of recent empirical articles in five prominent psychology journals, quantifying how often interactions are tested, how often the outcome family and link function are made explicit, and which response formats predominate. The second develops the link function problem in three common response formats: forced-choice accuracy with a chance floor, sum scores, and within-family link choices such as logit versus probit. Using simulations, it quantifies how often an inadequate link turns a purely additive structure into a statistically significant interaction, and whether standard diagnostics detect the misspecification. Table 4 summarizes these insights into a practical workflow for applied research.

Technical details are reported in Supplement A; a broader precomputed sensitivity atlas varying sample size, main-effect magnitude, scale location, and selected design parameters is reported in Supplement B.

Current Practice in Psychological Interaction Testing: A Preregistered Review

The review asks whether the scale-dependence problem is visible in current reporting practice: how often psychological articles test interactions, and how often they make the outcome family and link function explicit enough for a reader to know which interaction claim was tested. To our knowledge, it is the first review to quantify link-function reporting in psychological interaction tests. The coding is deliberately descriptive, based only on what a published report makes observable, so it documents reporting practice rather than the correctness of any substantive conclusion.

Review Method

The protocol was preregistered before any record was screened (<https://osf.io/bdu73/>), together with the coding data dictionary. The essentials are given here; the full protocol, sampling frame, and reliability are in Supplement A.

Following the preregistration, we sampled one journal from each of five areas: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). The five were selected on preregistered discretionary criteria: a first-quartile 2023 Impact Factor, a broad rather than narrowly specialized scope, not primarily publishing reviews, and strong reputation in their field, as jointly judged by the coauthors. From each journal's full set of 2024 articles in Scopus, records were screened in a fixed random order until 40 eligible interaction-testing articles were reached, for a preregistered total of 200. Because screening stopped once 40 interaction-testing articles were reached, the number of records screened, and therefore the denominator of the eligibility proportion, varied across journals. Eligible articles were empirical reports of original research; reviews, meta-analyses, editorials, theoretical and qualitative-only articles, replication-only studies, and psychometric validation studies were excluded. Detailed coding was applied only to articles testing at least one interaction, and only to observed-outcome analyses: articles whose relevant tests were purely latent-variable or structural-equation models fell outside the coding scheme (12 such exclusions).

Each interaction-testing article was independently coded by two researchers, with disagreements resolved by discussion or by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate passages within an article; all coding used in the review is human.

For each article, the coders recorded five things: whether at least one tested interaction used a non-identity link; whether the link function was explicitly reported; whether at least one interaction applied an incorrect identity-link analysis (i.e. applied it to a formally constrained observed outcome); whether at least one such interaction was statistically significant, or interpreted analogously in a Bayesian analysis; and which response-variable types were analyzed. By a formally incorrect identity-link analysis we mean a Gaussian linear regression with an identity link applied to an outcome with formal constraints, such as bounds, discreteness, positivity, or a known chance level. This is a conservative screening flag, applied only when the constraint was identifiable from the report; it is not an error rate and does not imply the substantive conclusion is wrong. The label is defined relative to the constrained process that generates the outcome (if the observed score were explicitly justified as the target metric, identity-link additivity would instead be a defensible modeling choice, a distinction we develop in the following sections). The full operational rules the coders followed, including borderline cases, are tabulated in Supplement A. Interrater agreement on the double-coded records was high (Cohen's κ between 0.66 and 0.81); per-variable agreement and coder margins are reported in Supplement A. Preregistered variable definitions, operational coding rules, variable-specific agreement, and by-journal results are also reported in Supplement A, Section S2.

Review Findings

First, interaction testing was common. Screening yielded 331 eligible empirical articles, of which 200 tested at least one interaction, 60.4 %. Under the stopping rule this proportion describes the records screened rather than a fixed-denominator sample, so we report it descriptively and treat the per-journal proportions as the more informative summary: these ranged from 38.1 % in Psychological Medicine to 85.1 % in Journal of Experimental Psychology:

General. Interaction testing was routine in all five journals.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated.

Even the 46 explicit cases rest on a rule looser than the preregistered one, which required a declaration of the link and treated naming a family or a generalized linear model as insufficient. In coding we also accepted link-specific model names, because in practice they identify the link: “logistic regression” names the logit and “probit regression” the probit. We also accepted “Poisson regression”, which names a family whose standard link is the log. The stricter preregistered rule would give a lower count.

Third, constrained outcomes were often analyzed on the identity scale. Among the 200 interaction-testing articles, 144 included at least one formally incorrect identity-link analysis of a constrained observed outcome, 72%, 95% CI [65.4, 77.8]. As noted, this is a conservative screening count, flagged only when the constraint was clear from the report. Of these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. This is not a comparison, because significance was coded only within the flagged analyses; that is, 62.0% of all interaction-testing articles combined a constrained outcome, an identity-link analysis, and a significant interaction claim. This is the same concern raised by work on outcome-model mismatch in interaction testing ([Domingue et al., 2024](#); [Liddell & Kruschke, 2018](#)). Table 1 summarizes these results.

The pattern was present across the sampled journals. In Supplement A, the proportion of formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect

Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric. The interval on the eligibility row is reported for completeness only: screening stopped at 40 interaction-testing articles per journal, denominator was not fixed by design.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

identity-link cases.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). This heterogeneity matters because the identity link has different implications across outcome types. Together, binary, accuracy, or proportion outcomes, sum scores or composites, and ordinal, Likert,

or rating outcomes appeared in 152 of the 200 articles (76.0%). The worked examples therefore focus on binomial and aggregated responses, including sums of binary or ordinal items. We thus develop simulation scenarios to show the consequences of link misspecification: many observed psychological variables are built from binary or ordered response events (be them proportions, ratings, averages, totals, or composites). For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Counts / error counts	9	200	4.5%
Neural / physiological measures	23	200	11.5%
Correlations / associations	8	200	4.0%
Difference / distance scores	17	200	8.5%

What the Present Review Shows, What It Does Not

As the first review to quantify link-function reporting in psychological interaction tests, it suggests three points. First, interaction testing is pervasive, in every journal sampled and in 60% of the eligible empirical articles screened. Second, the scale of additivity is usually left implicit: only 23% named the link function, and even that count leans on counting model names (such as

“logistic regression”) rather than the link function per se. Third, constrained outcomes are routinely analyzed on the identity scale: 72% of interaction-testing articles applied an identity-link analysis to an outcome with formal constraints, and most of these reported a significant interaction. In leading journals, then, interaction tests are common while the scale on which no interaction is assumed is rarely stated.

The review documents how analyses were reported. A descriptive review cannot establish how often published interactions are scale artifacts, because that judgment requires knowing the scale on which the data-generating process is additive. The remainder of the paper therefore turns to settings where the generating process is known by construction.

The Link Function Problem in Three Common Cases

To demonstrate the consequences of link misspecification we conduct three simulation studies:

1. *Forced-Choice Accuracy*: This case holds the binomial outcome family fixed and evaluates whether the link function must explicitly encode a task-implied chance floor (e.g., a 0.50 guessing threshold).
2. *Within-Family Link Choice (Logit vs. Probit)*: This case holds both the binary family and the 0-to-1 bounds fixed, testing whether choosing between these two standard link functions alters the results.
3. *Sum Scores*: This case adds the target metric as a third choice alongside family and link. A total built from item responses is itself a measurement mapping, so an analysis of the manifest score tests additivity in raw-score units, even though the substantive claim is generally made about the underlying construct.

Each case develops the conceptual problem and then reports a simulation that quantifies it. The three simulations ask whether a purely additive data-generating process on one scale is detected as a statistically significant interaction when modeled on another scale. A final simulation, reported in the next section, examines whether standard diagnostics reliably identify

the misspecified link. The simulation scenarios were chosen to represent plausible psychological data patterns. Each reproduces a common interaction-testing situation: two main effects that are well established and pretty large, for example a group difference combined with a condition effect or a developmental trend, and the question of whether one effect moderates the other. The two main effects were always tuned strong enough to push expected values close to a floor, ceiling, or chance level while keeping them clearly inside the observed range. This is the region where link curvature acts most strongly.

Each simulation applies the same design. Data are generated with no product term on the generating scale, and models assuming additivity on that scale and on plausible alternative scales are fitted to the same data. Every fitted model yields a product-term rejection rate. Rejection under a matched generating-scale model is the nominal rejection rate, whereas rejection under a mismatched-scale model is the pseudo-interaction detection rate.

Forced-Choice Accuracy and the Non-Zero Chance Floor

If a participant answers Y_i of k_i trials correctly, a binomial sampling model, $Y_i \sim \text{Binomial}(k_i, p_i)$, is the natural starting point. The family fixes the sampling process, but not the lower bound that is meaningful for the task. In an m -alternative forced-choice task, guessing yields an expected accuracy of $c = 1/m$, which is .50 in a two-alternative or true/false task, so a claim about performance above guessing places the lower asymptote of the performance curve at c , not at 0.

Three specifications treat this floor differently. A Gaussian identity model on the raw proportions ignores the trial process, assumes additivity in percentage-correct units, and can predict outside the 0-to-1 range. A standard binomial logit or probit model, $p_i = F(\eta_i)$ with F the logistic or standard-normal CDF, respects the sampling process and the 0-to-1 bounds and places the lower asymptote at 0. A chance-corrected binomial link keeps the binomial family and maps the linear predictor onto the interval from chance to 1,

$$p_i = c + (1 - c)F(\eta_i),$$

which for $c = .50$ becomes $p_i = .50 + .50 F(\eta_i)$; equivalently, the link acts on the above-chance quantity $(p_i - c)/(1 - c)$. A interaction term in the standard model tests additivity over the full 0-to-1 range, and a interaction term in the chance-corrected model tests additivity on a performance-above-chance scale that already encodes the task floor.

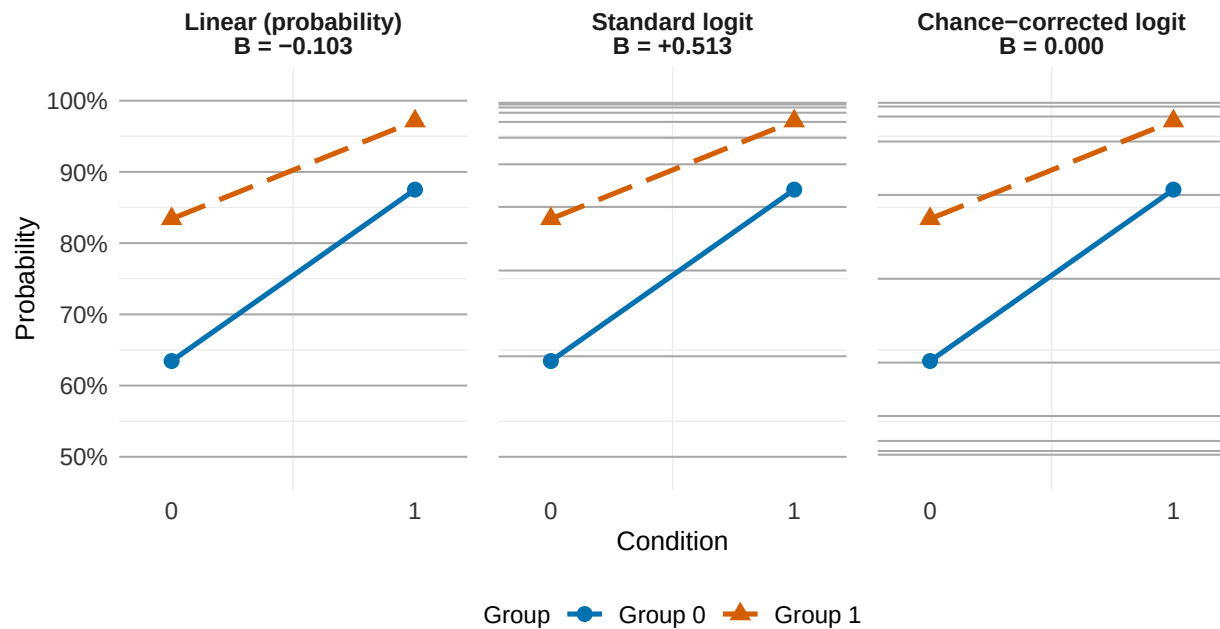
The mismatch is most severe when one group or condition lies near chance. Suppose age has the same effect in two groups on the performance-above-chance scale, with no true age-by-group interaction term there. Near .50 the chance-corrected curve has little response-scale room, so the lower-performing group shows stronger compression on the observed accuracy scale. A standard logit or probit link puts that compression near 0 in place of near .50. The age-by-group difference between differences is exactly zero on the chance-corrected scale and systematically nonzero on the standard logit scale, so the standard model detects a genuine property of its own scale (Figure 2).

The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c . Lapses, response biases, item heterogeneity, changes across trials, or systematic below-chance performance may instead require richer psychometric solutions such as item-response models (e.g., [Embretson & Reise, 2000](#); [Wichmann & Hill, 2001](#)).

The chance-corrected GLM link used here closely relates to psychometric-function models ([Knoblauch & Maloney, 2012](#); [Wichmann & Hill, 2001](#)) that estimate the chance process at the level where it actually operates. For example, the 3-parameter logistic Item Response Theory model includes an item-specific guessing parameter rather than a single chance level fixed at the design level ([Embretson & Reise, 2000](#)). The GLM link should thus be seen as a design-level approximation, not a substitute for item-level measurement models when the data and scientific question call for that additional structure.

Figure 2

A minimal 2×2 design crosses a condition (x-axis) with a group (color), shown on three scales. The four cell probabilities are identical in every panel; only the scale on which the interaction is evaluated changes. The data were generated with no condition-by-group interaction on the chance-corrected logit scale, which has a 0.50 chance floor, so the interaction coefficient is exactly zero on that scale. The same cells imply a positive interaction on the standard logit scale and a negative interaction on the raw probability scale. Light horizontal lines are equally spaced on each panel's own link scale, so equal steps on that scale map to unequal steps on the probability axis.



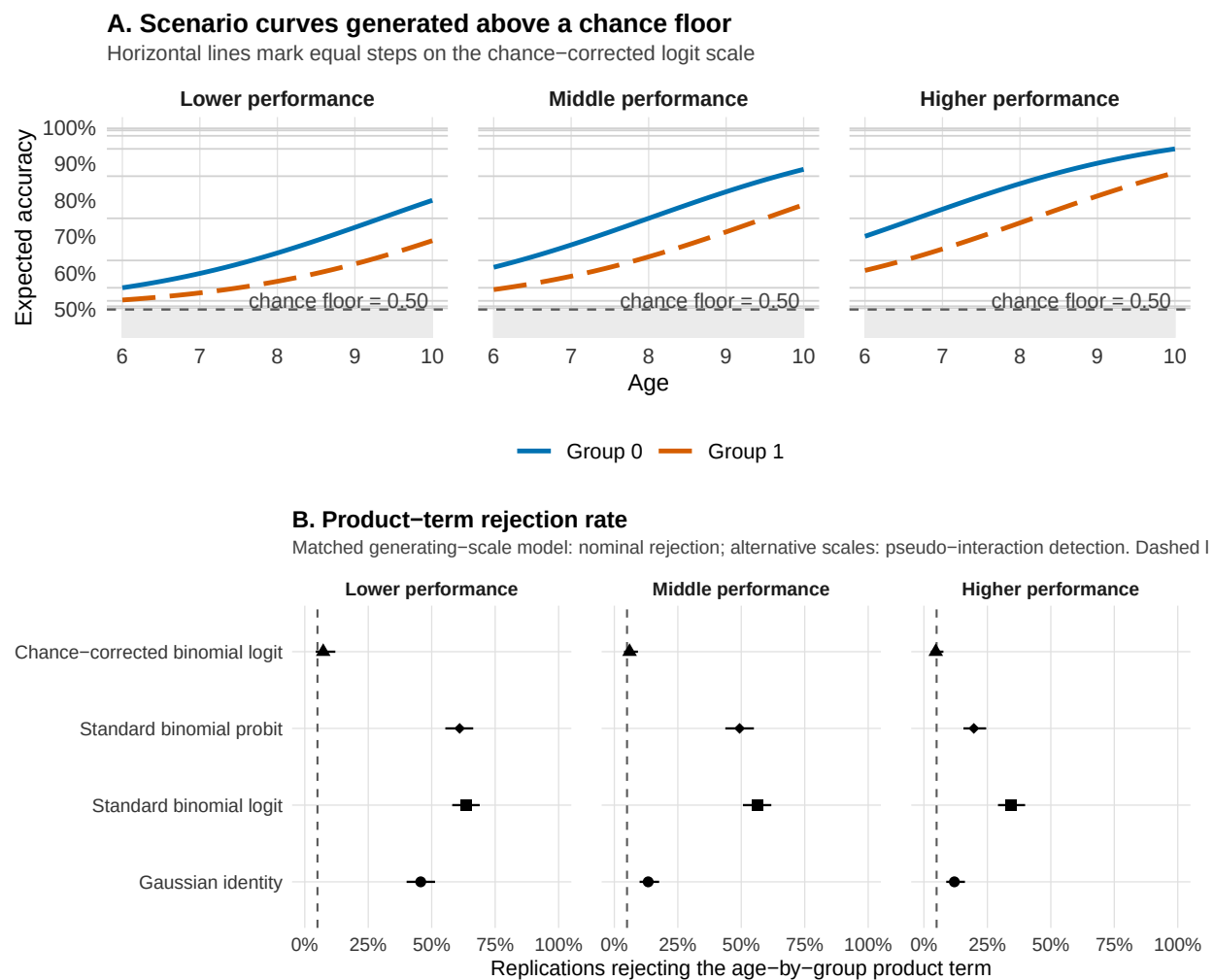
Simulation 1: Design and Results

In each of 300 replications we generated accuracy for $N = 250$ participants with $k = 20$ trials each, from a chance-corrected logit model with a .50 floor, a positive age effect, a group difference, and no age-by-group product term on the generating scale. Three scenarios differ only in overall performance, set by shifting the intercept: in the *lower-performance* scenario the two groups lie close to the .50 floor (expected accuracies roughly .53 to .80 across ages 6 to 10), in the *middle-performance* scenario they occupy the middle of the above-chance range (roughly .55 to

.88), and in the *higher-performance* scenario they approach the ceiling (roughly .61 to .94). Each simulated dataset was analyzed with four aggregate models, one row per participant: a Gaussian linear model on the observed proportions, standard binomial-logit and binomial-probit GLMs on the correct-response counts, and the chance-corrected binomial-logit GLM that matches the generating scale. We recorded how often each product-term test rejected at $\alpha = .05$. The matching model supplies the nominal rejection rate, while the three alternative scales supply pseudo-interaction detection rates. Figure 3 shows the simulation results; the per-scenario rates and the convergence counts are tabulated in Supplement A.

Figure 3

Simulation 1: forced-choice accuracy with a chance floor. Data are generated with no age-by-group product term on the chance-corrected logit scale. Panel A shows the resulting accuracy curves across three performance levels; the grey region lies below the .50 chance floor. Panel B shows product-term rejection rates by fitted model and scenario. The matched chance-corrected model remains near nominal alpha, whereas the alternative scales produce pseudo-interactions. The dashed line marks $\alpha = .05$.



The chance-corrected model stays near the nominal .05 in every scenario, with nominal rejection rates between 5% and 7%, because it matches the generating scale. Pseudo-interaction detection is highest where performance lies near the floor: in the lower-performance scenario the

standard logit rejects 64% of the time, the standard probit 61%, and the Gaussian identity 46%, against the nominal 5%. Detection falls as performance moves away from the floor, reaching 34% for the logit and 20% for the probit in the higher-performance scenario. When the true floor is .50 and the fitted link places it at 0, the product term compensates for the wrong no-interaction baseline. The same qualitative sensitivity to the chance floor also appears at chance levels 0.25 and 0.33 (Supplement B).

The chance-corrected model can be numerically demanding. Near the floor the likelihood carries little information about changes on the above-chance scale, and in the lower-performance scenario only 179 of the 300 replications returned a usable interaction test. Its rejection rate is therefore computed on the replications that converged, and these are not a random subset, since failures concentrate where the data are least informative about above-chance differences. Supplement A repeats all four models on the replications in which every model converged, and the ordering of the rates is unchanged. The instability is itself informative, signaling that the design gives limited evidence about above-guessing differences in that region. Weakly informative priors can help stabilize estimation when binomial data provide limited information ([Gelman et al., 2008](#)), although the added assumptions should be stated and checked.

Within-Family Link Choice

The forced-choice case involved a link that was plainly wrong for the task, because it placed the lower asymptote at 0 when the design fixed it at chance. But the link-function problem can also arise in less obvious cases. Logit and probit are both standard for binary data, respect the same 0-to-1 bounds, and usually produce very similar fitted probabilities. Their choice is therefore often guided by convention, ease of interpretation, or implementation, with little expected impact on substantive conclusions ([Agresti, 2002](#); [Yu & Huang, 2019](#)). This makes them a special case for illustrating the problem, as even two virtually interchangeable links define slightly different scales of additivity. Equal differences under one link are not exactly equal under the other, especially toward the tails, and this alone can alter the estimated product term.

Both links can be represented as latent-threshold models. They differ in the assumed latent

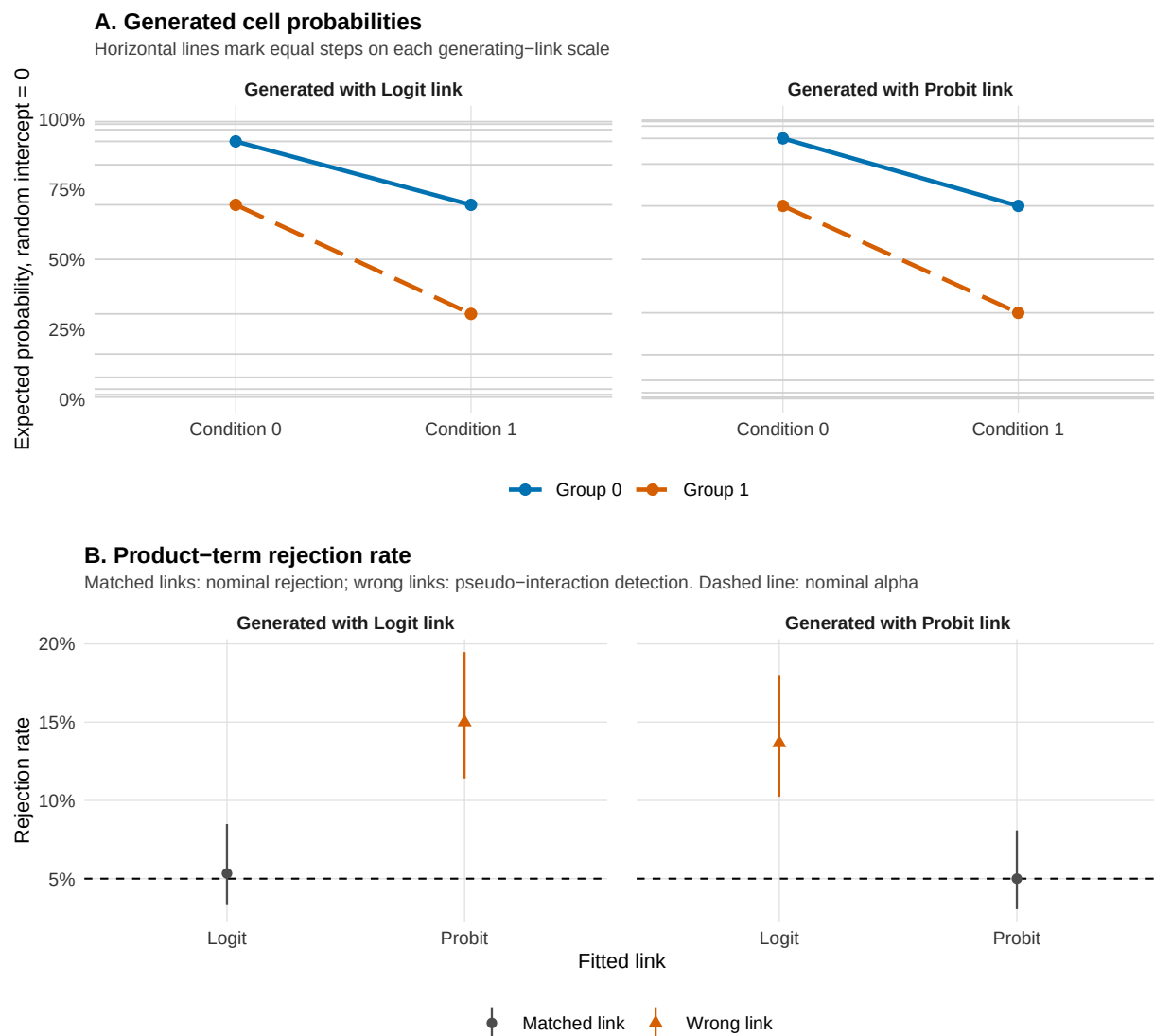
error distribution, logistic for the logit link and standard normal for the probit link, and therefore in curvature and tail behavior. The logit parameterization also gives coefficients a direct log-odds interpretation, whereas probit coefficients are expressed on a standard-normal latent scale, making it a choice of election when modeling psychological quantities such as ability, signal strength, response tendency, etc. (Dunn & Smyth, 2018).

Simulation 2: Design and Results

In each of 300 replications we generated repeated binary responses for $N = 700$ participants in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept (latent intraclass correlation 0.30). The quite large sample was chosen to make the relatively small systematic discrepancy between logit and probit links detectable, rather than to represent a typical psychological study. The condition-by-group interaction term was zero on the generating link scale. We crossed the generating link with the fitted link, using logit and probit in both roles, and fitted every model as a random-intercept GLMM, recording how often each returned a significant condition-by-group interaction term at $\alpha = .05$. Figure 4 shows the results; the full crossed-link rates are tabulated in Supplement A.

Figure 4

Simulation 2: logit versus probit in repeated-trial random-intercept GLMMs. Data are generated with no condition-by-group product term on the generating link scale. Panel A shows the corresponding cell probabilities under logit and probit links. Panel B shows product-term rejection rates for each generating–fitted link combination. Matched links remain near nominal alpha, whereas wrong links produce pseudo-interactions. Error bars are Wilson 95% confidence intervals; the dashed line marks $\alpha = .05$.



When the fitted link matches the generating link, the nominal rejection rate stays near .05:

5% for logit fitted to logit-generated data and 5% for probit fitted to probit-generated data.

Crossing the links raises the pseudo-interaction detection rate to 14% when a logit model is fitted to probit-generated data and 15% when a probit model is fitted to logit-generated data, roughly three times the nominal rate. The wrong-link detection rates are smaller than in the forced-choice case, and they appear among links that share the same family, the same 0-to-1 bounds, and nearly overlapping fitted curves in the middle of the probability range, the setting many applied analysts treat as interchangeable. Under the wrong link the same cell probabilities imply a small but deterministic nonzero product term on the fitted scale. Supplement B shows that its detectability varies non-monotonically with latent location, becoming small in some regions and much larger in others. A result that is stable across both links is easier to defend; one that appears only under a single link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links. The deterministic product term induced by evaluating each generating model on the alternative link scale is reported in Supplement A, Section S5.

Sum Scores

Many psychological outcomes are computed as total sum scores across questionnaire items, task items, symptoms, errors, or ratings. In the two cases above the outcome was a directly observed response, and only the link was in question. A sum score is different, because the score itself is already the result of an analytic choice. What is observed are the individual item responses, which are binary or ordinal; the total is a summary of them, produced by a scoring rule that gives every item the same weight and every point of the scale the same width. In psychological research the construct of interest is often the underlying, generally continuous dimension, so the total can be seen as a remapped version of that continuum. The link question then arrives together with a prior question about which metric the claim is about.

That remapping is not linear. The bounds are crucial here, because the construct may have no floor or ceiling while the observed total score has both. The mapping is best read as an exchange rate between the two scales, that is, how much of the latent dimension is needed to gain one raw-score point. That rate is not constant. Near the middle of the scale a small latent shift is

enough to move the total by a point, while near either bound a wide stretch of the latent dimension is compressed into the last available point, because few items remain that the respondent can still change. To illustrate, with nine items scored 0-3, moving from 13 to 14 points requires less change on the latent dimension than moving from 26 to 27. A constant latent effect therefore produces smaller raw-score differences as the total approaches an end, and the same reasoning applies wherever the instrument is unevenly informative along the latent dimension, so compression is not confined to the endpoints.

Bounded, discrete, and skewed distributions are common in psychological data ([Micceri, 1989](#)), and the consequences of analysing ordinal responses as if they were metric have been documented in detail ([Liddell & Kruschke, 2018](#)). Fitting a Gaussian identity model to a total assumes instead that a predictor moves the score by the same number of points at every level of the other predictor. A group/condition sitting near a bound moves by fewer points even when the data-generating process on the latent dimension is the same in both groups/conditions. Behaviour-genetic research has met the same problem when testing whether genetic effects vary across environments, since those interactions are usually tested on summed questionnaire scores: analysed at the item level, the same data show that the total can conceal real interactions or present spurious ones through properties of the observed score ([Molenaar & Dolan, 2014](#)). This is the sum-score form of outcome-model misspecification and target-metric mismatch ([Domingue et al., 2024](#)).

The problem we raise sits next to a recent, wider debate about whether sum scores should be replaced by factor scores or latent-variable models. Some argue that sum scores are transparent, reliable enough for many purposes, comparable across studies, and tied to how instruments are scored in applied settings ([Widaman & Revelle, 2023](#)); the counterargument is that a total encodes strong and usually untested assumptions about how items represent the construct ([McNeish, 2023](#); [McNeish & Wolf, 2020](#)). While we do not enter the debate, we add a narrower point that was left aside: a model of the manifest total defines the absence of an interaction in raw-score points, so an interaction found there is an interaction on the score, and it becomes a claim about the construct

only under the additional assumption that the two are linearly related.

When item-level data are available and the claim concerns latent constructs, an explicit measurement model such as SEM, item-level ordinal regression, or IRT-inspired specifications could be preferred, since it addresses the mapping directly, though at the price of moving the assumptions into the measurement model (Bürkner & Vuorre, 2019). Often, however, researchers deliberately work with the total score for the practical reasons listed above. An option is to set a family that respects the bounds and the discreteness, for example modelling the total as a bounded count with a beta-binomial likelihood (Wilcox, 1981). However, we suggest a deliberately unusual option that moves only the link: keep the observed total score under a Gaussian likelihood, but rescale the score to the proportion of the maximum and use a probit link function, as in Simulation 3 below. This is an *ad hoc* device, not a standard practice, but it illustrates whether the interaction conclusion changes if additivity is placed on a nonlinear scale that compresses the expected values toward the score bounds. It remains a model of the total, it will not recover the construct, and the Gaussian likelihood is still misspecified in its variance, since a bounded score has a variance that depends on its mean: what the probit link corrects is only the scale on which additivity is defined, not the distributional form. Read against the sum-score debate, it is a precaution for researchers who have decided to analyse totals anyway, but acknowledge that the interaction still needs a scale on which additivity is plausible.

Simulation 3: Design and Results

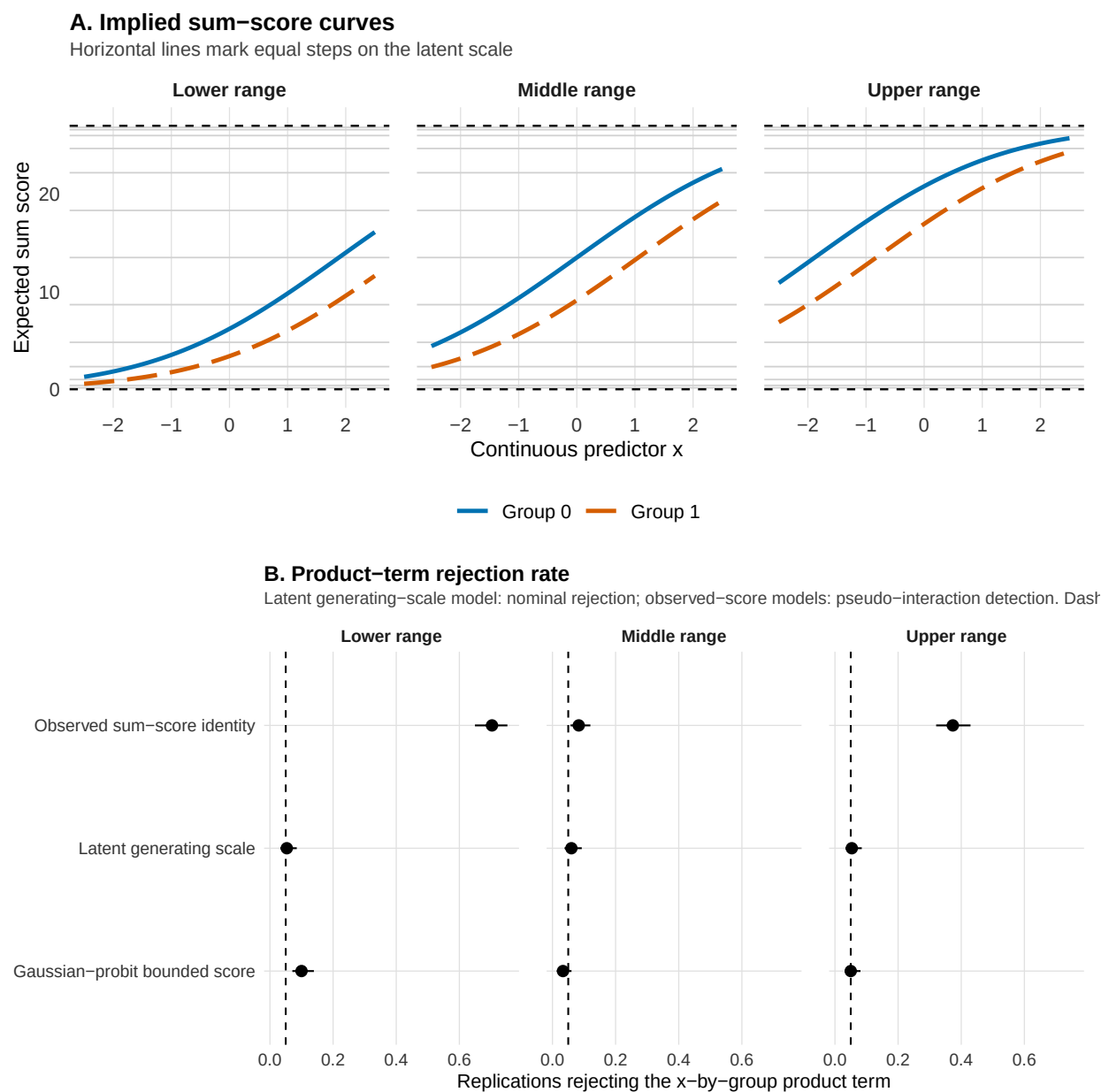
The bounded-score probit model was compared with two natural alternatives: the observed total and the latent scale it approximates. In each of 300 replications, we generated data for $N = 600$ participants from a continuous latent variable with a continuous predictor x , a group effect, and no x -by-group interaction. The latent values generated 9 ordinal items scored 0 to 3, which were summed to a total ranging from 0 to 27. Item thresholds were shifted across three scenarios so that expected totals fell in the middle, lower, or upper part of the score range.

We analyzed each total in three ways: a Gaussian identity model of the observed sum score, a Gaussian-probit bounded-score model, and the true latent generating scale. The

bounded-score model applies a probit link to a continuity-corrected score proportion and rescales predictions to the original sum-score metric. We used probit because its standard-normal linear predictor provides a simple correspondence with the latent scale. This choice is still substantive, as shown in the previous section. The latent model serves as an idealized benchmark available only in simulation. We recorded the proportion of significant x-by-group interactions at $\alpha = .05$. Results are shown in Figure 5, with per-scenario rates in Supplement A, Section S4.

Figure 5

Simulation 3: sum scores as bounded and discrete outcomes. Data are generated with no x -by-group product term on the latent scale. Panel A shows how the same additive latent structure maps onto sum scores in the middle, lower, and upper parts of the score range. Panel B shows product-term rejection rates for the latent benchmark and two observed-score models. The latent model remains near nominal alpha, whereas the observed-score models can produce pseudo-interactions. The dashed line marks $\alpha = .05$. The latent benchmark is available only in simulation.



The latent benchmark remained close to the nominal .05 rejection rate in every scenario, from 5% to 6%. The identity model behaved similarly only in the middle of the score range, where the sum score was approximately linear in the latent variable (8%). Rejection increased as expected totals moved toward a bound, reaching 37% in the upper range and 70% in the lower range. Compression of the latent scale makes the same latent group difference appear to change across levels of the predictor. The bounded-score probit model remained close to nominal alpha across scenarios, from 3% to 10%. It respects the score bounds, although it is still a model of the manifest total.

This pattern mainly depends on where observations fall within the score range. Across the broader scenarios in Supplement B, scale location affected identity-model rejection much more than sample size, item count, or latent dispersion. Raw-score additivity can approximate latent-scale additivity when scores remain away from the bounds. Compression near a bound can instead create an apparent interaction from a constant latent effect. The bounded-score probit model can therefore serve as a useful sensitivity analysis. For claims about interactions between latent constructs, an explicit measurement model provides a more direct approach, for example SEM, item-level ordinal models, or IRT-inspired models. These approaches still depend on assumptions about thresholds, links, loadings, and latent-response scales, but they make those assumptions part of the interaction model.

Can Standard Diagnostics Catch a Wrong Link?

The simulations above assume the analyst does not question the fitted model. In practice, a researcher who finds an interaction may run model checks before reporting it. We examine three: a Pregibon-style added-term link check, simulation-based residual diagnostics, and information criteria. This set represents distinct strategies, probing the assumed link, screening for misfit of any kind, and ranking a set of candidate models. Such strategies are checks that applied psychologists might routinely apply. A further strategy, estimating the shape of the link from the data through a parametric family of links, is more direct and rarely used in psychology (Aranda-Ordaz, 1981; Czado & Santner, 1992; Stukel, 1988). None of the strategies directly

identifies the right scale on which the no-interaction claim is defined. At best, they point to the link that fits the data best among a set of alternatives.

A link test of the kind proposed by Pregibon targets link adequacy directly, in the added-variable form used here ([Pregibon, 1980](#)). The model is fitted, the fitted linear predictor $\hat{\eta}$ is extracted, and the model is refitted with $\hat{\eta}^2$ added as an extra predictor. If the link is adequate the squared term carries no additional information and should be non-significant; a significant term signals curvature that the current link leaves unexplained. The signal is not specific to the link: an omitted nonlinear term, an omitted covariate, or a missing interaction produce it as well. A non-significant result means only that this diagnostic did not detect a departure in this sample.

Residual diagnostics cover more ground. In a GLM, raw, Pearson, and deviance residuals have no common reference distribution across families, which makes residual plots difficult to read. Randomized quantile residuals tackle this problem via simulation ([Dunn & Smyth, 1996](#)): many datasets are generated from the fitted model, and each observed value is located within the distribution of its own simulated values, so a value at the median of that distribution receives .50 and a value in the far tail receives a value near 0 or 1. Under an adequate model these positions are uniform between 0 and 1 for every family, so misfit appears as departure from uniformity, as over- or underdispersion, or as residual structure over fitted values and predictors. We use the implementation in the DHARMA package of R ([Hartig, 2024](#)). The wider coverage comes at the cost of specificity, however: link misspecification, wrong family, missing nonlinearity, missing interaction, and target-metric mismatch all appear as the same kind of departure from uniformity.

Information criteria such as AIC, BIC, WAIC, and LOO compare a supplied set of models. For link choice, the relevant comparison keeps the response, family, and structural formula fixed and varies only the link. This is best viewed as a targeted sensitivity analysis. It has two important limits. First, the result is restricted to the candidate links that were fitted. Second, the interaction term is already present in each model and can absorb part of the curvature produced by the wrong link. A misspecified model may therefore fit almost as well as the target model while still placing additivity on the wrong scale. In our comparisons, the candidate models have the same number of

parameters, so their AIC ranking depends entirely on the log-likelihood.

Diagnostic Simulation

We simulate five scenarios covering different cases, each being tied to a previous example in the paper. The first is tied to the simple count example in Figure 1. Data are generated from a Poisson-log model in which expected errors decline with age, with no age-by-group interaction term on the log-count scale. The diagnostic scenario uses the same design as the figure, with its own coefficients recorded in the generated scenario table: intercept 2.20, age effect -0.55 per year after age 6, and group effect 0.45. The intercept and the group effect are slightly smaller than in the illustrative figure, which uses 2.60 and 0.65; the age effect is the same. The fitted model keeps the Poisson family and interaction formula but uses an identity link, making this a same-family comparison in which only the link changes. In real research scenarios, however, family and link often covary.

The second used the lower-performance forced-choice scenario from Simulation 1. Data were generated with a 0.50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group interaction on the chance-corrected logit scale. The fitted model was a standard binomial-logit interaction model. It retained the binomial trial structure but placed the lower asymptote at 0. The chance-corrected model was initialized from the standard-logit fit and optimized with BFGS, with Nelder–Mead as fallback when needed.

The third scenario reproduced the probit-generated, logit-fitted case from Simulation 2. Data consisted of repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. Fixed effects were intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition interaction. The fitted model was the corresponding random-intercept logit GLMM with the same formula.

The fourth scenario was a continuous-predictor version of the same binary case. It kept the probit generating link, logit fitted link, group effect, focal-predictor effect, random intercept, and latent ICC, while replacing the two-level condition with a continuous predictor sampled from 0 to 1. This allows residual checks that rely on variation along a continuous predictor.

The fifth scenario uses the Gamma response-time example from the introduction. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when $x = 0$ and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x -by-group interaction term. The fitted model keeps the Gamma family and the interaction formula and uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after obtaining a significant interaction. The question is therefore whether the same model that produces the pseudo-interaction also shows enough evidence of misspecification to warn the analyst. This is a demanding test, since the interaction term can already absorb part of that curvature.

The DHARMA workflow included four checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual structure over the focal predictor ([Dunn & Smyth, 1996](#); [Hartig, 2024](#)). For continuous predictors we used a quantile check; for the categorical 2×2 design we used a design-cell comparison. We also applied the Pregibon-style added-term link check ([Pregibon, 1980](#)). As a calibration check, the same DHARMA and Pregibon-style procedures were applied to the corresponding correctly specified interaction model in each replication.

AIC was used only for relative comparison among prespecified candidate mean models fitted to the same data with the same interaction formula, keeping family or sampling model fixed as described above and varying only the link. The chance-floor scenario compared a standard logit model with a chance-corrected logit model with a design-implied lower asymptote. We report both the proportion of replications in which AIC favored the target model and the distribution of ΔAIC , defined as AIC of the misspecified model minus AIC of the target model. Positive values therefore favor the target link. Because no inconclusive category was used, the complementary proportion favors the misspecified model when both values are available.

Table 3 shows that diagnostic sensitivity varied widely across scenarios. In the count case, the wrong link produced pseudo-interactions in 96% of replications. Pregibon detected the problem in 99%, while the most sensitive DHARMA check reached 88%. In the chance-floor case,

Table 3

Diagnostic simulation across five link-misspecification scenarios. Pseudo-interaction reports how often the misspecified model detected an interaction that was absent on the generating link scale. DHARMA gives the range across the four prespecified residual checks. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model, with the structural formula held fixed. The complementary proportion favors the misspecified model when both AIC values are available. CI = Wilson 95% confidence interval. 'N.A.' indicates that the Pregibon-style added term is not identifiable in the saturated 2times2 interaction model: the squared population-level linear predictor is exactly collinear with the existing fixed effects.

Case	Target model	Misspecified model	Pseudo-interaction rate	DHARMA	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.957 [0.927, 0.975]	0.000-0.883 (4)	0.986	0.986
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.587 [0.530, 0.641]	0.000-0.060 (4)	0.317	0.813
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.127 [0.094, 0.169]	0.000-0.013 (4)	N.A.	0.950
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.073 [0.049, 0.109]	0.000-0.003 (4)	0.043	0.937
Gamma mean response time	Gamma log	Gamma inverse	0.297 [0.248, 0.351]	0.000-0.063 (4)	0.340	0.793

the corresponding rates were 59% for the pseudo-interaction, 32% for Pregibon, and at most 6% for DHARMA. In the Gamma case, they were 30%, 34%, and at most 6%, respectively.

The logit-probit cases were harder to diagnose. In the saturated 2×2 design, pseudo-interactions occurred in 13% of replications and DHARMA detected at most 1%; Pregibon

was not identifiable. With a continuous predictor, the corresponding rates were 7% and 0%, while Pregibon detected 4%. AIC was more sensitive in both cases, favoring the target probit model in 95% and 94% of replications.

The chance-floor case also shows that better fit need not imply a large separation between models. AIC favored the chance-corrected model in 81% of replications, with a median difference of 2.4 AIC units. The two models were within two units in 39.0% (117/300) of finite comparisons. Among replications with a significant pseudo-interaction, all applicable DHARMA checks were non-significant in 93.8% (165/176) of replications.

Overall, the diagnostics were informative in some settings and weak in others, even when the pseudo-interaction significance rate is pretty high. Under the correct link, DHARMA flagged a problem in at most 4.7% of replications. Pregibon flagged at most 5.3% in the count, chance-floor, and Gamma scenarios, and 4.3% in the continuous-predictor GLMM. Its low sensitivity in the latter case therefore reflects limited detection of that form of misspecification, not poor calibration. Full calibration results are reported in Supplement A, and Supplement B shows how pseudo-interaction rates, Pregibon detection, and AIC preference change across sample size and scale location.

From Link Choice to Sensitivity Analysis

The practical implication of this framework is that an interaction analysis should make three choices explicit: the link that defines additivity, the metric the claim refers to, and the sensitivity checks used to assess that choice (Table 4). The observed metric may sometimes be the relevant target. For example, a two-point difference on a clinical or educational score may be the quantity that matters in practice, and additivity in raw-score units may then be reasonable. The key point is that this should be a substantive choice, not the by-product of using a default model. When possible, the link should be justified from theory, task structure, or measurement ([Baranger et al., 2023](#); [Castillo et al., 2026](#)).

When link choice can affect the interaction conclusion, we suggest that the main robustness check is to perform a sensitivity analysis across plausible links. Theory, task

structure, and measurement should still guide the preferred specification, and diagnostics can provide evidence about model fit. The key robustness question, however, remains whether the interaction conclusion survives across alternative plausible links. Researchers should therefore compare the same target response-scale contrast across alternatives and report whether the result is stable or link-conditional. If it changes, that dependence is itself part of the result (McCabe et al., 2022; Mize, 2019; Mustillo et al., 2018; Rohrer & Arel-Bundock, 2026).

Table 4

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary / proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert / ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score / composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, bounded-count, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts / error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time / skewed positive outcome	Lognormal, Gamma log/inverse, Inverse Gaussian, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

Discussion

The same data can support different interaction conclusions under different plausible model specifications. This is the familiar scale dependence of interactions (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). In GLMs, this involves the choice of outcome family, link function, and target metric. Our review suggests that these choices are often left implicit, while the simulations show that they can matter even when the fitted model appears broadly reasonable.

More specifically, our review showed that interaction tests are common, link functions are often implicit, and many interaction-testing articles use identity-link analyses on outcomes for which observed-scale additivity is not self-evident. The simulation studies showed the risk of link misspecification, even within an appropriate family, when two main effects are present. Forced-choice accuracy offers a very clear case: the binomial family is correct, but the standard canonical link function misplaces the lower asymptote. The logit-probit comparison then shows that the problem persists even when the response bounds are correctly specified, between links that look similar over most of the range. Sum scores extend it to the target metric itself, where the manifest score is already a measurement mapping from item responses, and where even a likelihood suited to a bounded count leaves the scale of additivity to be chosen.

These results extend previous work showing that interaction effects are often difficult to estimate precisely even when the model form is correct (Baranger et al., 2023; Castillo et al., 2026). Here, the difficulty is specific to link, which can produce statistically significant interaction terms even when the true interaction is zero on the generating scale. The diagnostic simulation suggests that ordinary post-fit checks of the interaction model reveal link-driven interactions unevenly. Residual checks, Pregibon-style tests, and information criteria were useful in some settings, but their performance depended on the candidate set. The chance-floor scenario is the clearest case: a wrong link produced a large pseudo-interaction while residual checks showed no sign of misfit. There the check that responded was the explicit comparison between the two candidate links, and it separated them by a margin much smaller than the distance between the two interaction conclusions. This is diagnostic misalignment, not uniform diagnostic failure, and

it suggests that diagnostics should be treated as evidence about model fit within a candidate set, not as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Three limitations bound the scope of the conclusions. First, the review was descriptive and limited to five reference journals across major areas of psychology. Its numerical estimates should therefore be read as reflecting reporting practice in high-standard outlets during the sampled year. A broader mapping across journals, years, and subfields could later support reporting guidance on interaction tests with constrained outcomes. Second, the review coded directly observable reporting features rather than judging, for each article, whether its interaction was scale-dependent enough to be removable. Finally, the simulations are mechanism demonstrations aimed at a specific regime, two established main effects that jointly expose observations to link-function curvature. Supplement B broadens these scenarios varying the data-generating parameters, but it still does not estimate prevalence or cover all real-life complications.

In conclusion, interaction effects are often tested while the scale of additivity remains implicit. Link-aware testing separates three questions that are often run together: which outcome family is plausible, which link defines the scale of additivity, and whether the interaction conclusion is stable across plausible alternatives. This makes the scope of the claim explicit. A result that is stable across defensible links is more robust. If it changes, the interaction should be reported as conditional on the chosen scale.

References

- Agresti, A. (2002). *Categorical data analysis* (2nd ed.). John Wiley & Sons.
- Ai, C., & Norton, E. C. (2003). Interaction terms in logit and probit models. *Economics Letters*, 80(1), 123–129. [https://doi.org/10.1016/S0165-1765\(03\)00032-6](https://doi.org/10.1016/S0165-1765(03)00032-6)
- Aranda-Ordaz, F. J. (1981). On two families of transformations to additivity for binary response data. *Biometrika*, 68(2), 357–363. <https://doi.org/10.1093/biomet/68.2.357>
- Baranger, D. A. A., Finsaas, M. C., Goldstein, B. L., Vize, C. E., Lynam, D. R., & Olino, T. M. (2023). Tutorial: Power analyses for interaction effects in cross-sectional regressions.

- Advances in Methods and Practices in Psychological Science*, 6(3), 10.1177/25152459231187531. <https://doi.org/10.1177/25152459231187531>
- Bürkner, P.-C., & Vuorre, M. (2019). Ordinal regression models in psychology: A tutorial. *Advances in Methods and Practices in Psychological Science*, 2(1), 77–101. <https://doi.org/10.1177/2515245918823199>
- Busemeyer, J. R., & Jones, L. E. (1983). Analysis of multiplicative combination rules when the causal variables are measured with error. *Psychological Bulletin*, 93(3), 549–562. <https://doi.org/10.1037/0033-2909.93.3.549>
- Castillo, A., Miller, J. D., Vize, C., Baranger, D. A. A., & Lynam, D. R. (2026). When do interaction/moderation effects stabilize in linear regression? *Advances in Methods and Practices in Psychological Science*, 9(1), 10.1177/25152459251407860. <https://doi.org/10.1177/25152459251407860>
- Cox, D. R. (1984). Interaction. *International Statistical Review / Revue Internationale de Statistique*, 52(1), 1–24. <https://doi.org/10.2307/1403235>
- Czado, C., & Santner, T. J. (1992). The effect of link misspecification on binary regression inference. *Journal of Statistical Planning and Inference*, 33(2), 213–231. [https://doi.org/10.1016/0378-3758\(92\)90069-5](https://doi.org/10.1016/0378-3758(92)90069-5)
- Domingue, B. W., Kanopka, K., Trejo, S., Rhemtulla, M., & Tucker-Drob, E. M. (2024). Ubiquitous bias and false discovery due to model misspecification in analysis of statistical interactions: The role of the outcome’s distribution and metric properties. *Psychological Methods*, 29(6), 1164–1179. <https://doi.org/10.1037/met0000532>
- Dunn, P. K., & Smyth, G. K. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics*, 5(3), 236–244. <https://doi.org/10.1080/10618600.1996.10474708>
- Dunn, P. K., & Smyth, G. K. (2018). *Generalized Linear Models With Examples in R*. Springer New York. <https://doi.org/10.1007/978-1-4419-0118-7>
- Embretson, S. E., & Reise, S. P. (2000). *Item response theory for psychologists*. Lawrence Erlbaum Associates. <https://doi.org/10.4324/9781410605269>

- Gelman, A., Jakulin, A., Pittau, M. G., & Su, Y.-S. (2008). A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics*, 2(4), 1360–1383. <https://doi.org/10.1214/08-AOAS191>
- Hartig, F. (2024). *DHARMa: Residual diagnostics for hierarchical (multi-level / mixed) regression models*. <https://doi.org/10.32614/CRAN.package.DHARMa>
- Knoblauch, K., & Maloney, L. T. (2012). *Modeling Psychophysical Data in R*. Springer. <https://doi.org/10.1007/978-1-4614-4475-6>
- Liddell, T. M., & Kruschke, J. K. (2018). Analyzing ordinal data with metric models: What could possibly go wrong? *Journal of Experimental Social Psychology*, 79, 328–348. <https://doi.org/10.1016/j.jesp.2018.08.009>
- Loftus, G. R. (1978). On interpretation of interactions. *Memory & Cognition*, 6(3), 312–319. <https://doi.org/10.3758/BF03197461>
- Lubinski, D., & Humphreys, L. G. (1990). Assessing spurious "moderator effects": Illustrated substantively with the hypothesized ("synergistic") relation between spatial and mathematical ability. *Psychological Bulletin*, 107(3), 385–393. <https://doi.org/10.1037/0033-2909.107.3.385>
- McCabe, C. J., Halvorson, M. A., King, K. M., Cao, X., & Kim, D. S. (2022). Interpreting interaction effects in generalized linear models of nonlinear probabilities and counts. *Multivariate Behavioral Research*, 57(2-3), 243–263. <https://doi.org/10.1080/00273171.2020.1868966>
- McCullagh, P., & Nelder, J. A. (1989). *Generalized linear models* (2nd ed.). Chapman; Hall.
- McNeish, D. (2023). Psychometric properties of sum scores and factor scores differ even when their correlation is 0.98: A response to Widaman and Revelle. *Behavior Research Methods*, 55(8), 4269–4290. <https://doi.org/10.3758/s13428-022-02016-x>
- McNeish, D., & Wolf, M. G. (2020). Thinking twice about sum scores. *Behavior Research Methods*, 52(6), 2287–2305. <https://doi.org/10.3758/s13428-020-01398-0>
- Micceri, T. (1989). The unicorn, the normal curve, and other improbable creatures. *Psychological*

Bulletin, 105(1), 156–166.

Mize, T. D. (2019). Best practices for estimating, interpreting, and presenting nonlinear interaction effects. *Sociological Science*, 6, 81–117. <https://doi.org/10.15195/v6.a4>

Molenaar, D., & Dolan, C. V. (2014). Testing systematic genotype by environment interactions using item level data. *Behavior Genetics*, 44(3), 212–231. <https://doi.org/10.1007/s10519-014-9647-9>

Mustillo, S. A., Lizardo, O. A., & McVeigh, R. M. (2018). Editors' Comment: A Few Guidelines for Quantitative Submissions. *American Sociological Review*, 83(6), 1281–1283. <https://doi.org/10.1177/0003122418806282>

Nelder, J. A., & Wedderburn, R. W. M. (1972). Generalized linear models. *Journal of the Royal Statistical Society: Series A (General)*, 135(3), 370–384. <https://doi.org/10.2307/2344614>

Pregibon, D. (1980). Goodness of link tests for generalized linear models. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 29(1), 15–24. <https://doi.org/10.2307/2346405>

Rimpler, A., Kiers, H. A. L., & Ravenzwaaij, D. van. (2026). Anything Goes: Statistical Interactions Without Substantive Theory. *Computational Brain & Behavior*. <https://doi.org/10.1007/s42113-026-00305-8>

Rohrer, J. M., & Arel-Bundock, V. (2026). Models as Prediction Machines: How to Convert Confusing Coefficients Into Clear Quantities. *Advances in Methods and Practices in Psychological Science*, 9(2), 25152459261424825. <https://doi.org/10.1177/25152459261424825>

Rohrer, J. M., & Arslan, R. C. (2021). Precise Answers to Vague Questions: Issues With Interactions. *Advances in Methods and Practices in Psychological Science*, 4(2), 25152459211007368. <https://doi.org/10.1177/25152459211007368>

Stukel, T. A. (1988). Generalized logistic models. *Journal of the American Statistical Association*, 83(402), 426–431. <https://doi.org/10.1080/01621459.1988.10478613>

Wagenmakers, E.-J., Krypotos, A.-M., Criss, A. H., & Iverson, G. (2012). On the interpretation of removable interactions: A survey of the field 33 years after Loftus. *Memory & Cognition*,

40(2), 145–160. <https://doi.org/10.3758/s13421-011-0158-0>

Wichmann, F. A., & Hill, N. J. (2001). The psychometric function: I. Fitting, sampling, and goodness of fit. *Perception & Psychophysics*, 63(8), 1293–1313.

<https://doi.org/10.3758/BF03194544>

Widaman, K. F., & Revelle, W. (2023). Thinking thrice about sum scores, and then some more about measurement and analysis. *Behavior Research Methods*, 55(2), 788–806.

<https://doi.org/10.3758/s13428-022-01849-w>

Wilcox, R. R. (1981). A review of the beta-binomial model and its extensions. *Journal of Educational Statistics*, 6(1), 3–32. <https://doi.org/10.2307/1165046>

Yu, S., & Huang, X. (2019). Link misspecification in generalized linear mixed models with a random intercept for binary responses. *TEST*, 28(3), 827–843.

<https://doi.org/10.1007/s11749-018-0602-6>